

III.7 Fotonový plyn

→ fotony: Coulombova kalibrace + periodická o.p.

⇒ standardní procedura s disperzní relací $E = pc$

$$\Gamma(E) \longrightarrow \frac{V}{h^3} g \int_{\varepsilon(p) \leq E} d^3p = 2 \frac{V}{h^3} \frac{4}{3} \pi \left(\frac{E}{c}\right)^3$$

↑ 2 polarizace

$$\gamma(E) = \frac{d\Gamma(E)}{dE} = \frac{V}{h^3} \frac{8\pi}{c^3} E^2$$

→ fotony: nezachovávají počet částic ⇒ $\mu = 0$

→ hustota:

$$\begin{aligned} \rho &= \frac{1}{V} \int_0^\infty dE \gamma(E) f^+(E) = \frac{8\pi}{(hc)^3} \int_0^\infty \frac{E^2}{e^{\beta E} - 1} dE \\ &= \frac{8\pi}{(hc)^3} \frac{1}{\beta^3} \int_0^\infty \frac{x^2}{e^x - 1} = \frac{8\pi}{(hc)^3} \zeta(3) \underbrace{\Gamma(3)}_2 (k_B T)^3 \end{aligned}$$

$$\rho = \frac{16\pi}{(hc)^3} \zeta(3) (k_B T)^3$$

→ tlak

$$\begin{aligned} p &= \frac{1}{V} \int_0^\infty dE \Gamma(E) f^+(E) = \frac{8\pi}{3(hc)^3} \int_0^\infty dE \frac{E^3}{e^{\beta E} - 1} \\ &= \frac{8\pi}{3(hc)^3} \frac{1}{\beta^4} \int_0^\infty \frac{x^3}{e^x - 1} = \frac{8\pi}{3(hc)^3} \zeta(4) \Gamma(4) (k_B T)^4 \end{aligned}$$

$$p = \frac{16\pi}{(hc)^3} \zeta(4) (k_B T)^4$$

→ vnitřní energie ⇒ třikrát $\Gamma(E) \sim E^3 \Rightarrow u = 3p$

$$u = \frac{48\pi}{(hc)^3} \zeta(4) (k_B T)^4$$